DECODELABS

Project 2: Exploratory Data Analysis

Understanding order patterns, product performance, customer activity and order values

Dataset: DecodeLabs cleaned order dataset
Records: 1,200 | **Columns:** 14

1. Executive Summary

This project applies Exploratory Data Analysis (EDA) to a cleaned e-commerce order dataset. The purpose is to understand the dataset's structure, summarize important numerical measures, explore patterns across products and order statuses, examine monthly activity, and identify unusually high order values.

Key results: The dataset contains 1,200 orders from 1 January 2023 to 30 June 2025. There are 1,189 unique customers. The average order value is 1,053.97, while the median is 823.62. The total of the recorded order values is 1,264,761.96 across all statuses.

Important interpretation note: the total order value includes Cancelled, Returned, Pending, Shipped and Delivered records. It is therefore described as **total recorded order value**, not recognized revenue or completed sales.

2. Project Objectives

• Understand the structure and scope of the cleaned dataset.
• Calculate count, mean and median for relevant measures.
• Examine distributions and compare categories.
• Identify monthly patterns and unusually high values.
• Summarize observations and state limitations clearly.

3. Dataset Overview

The dataset contains 1,200 rows and 14 columns. Each row represents one order. Fields include order ID, date, customer ID, product, quantity, unit price, payment method, order status, items in cart, coupon code, referral source and total price.

Dataset property	Result
Rows / orders	1,200
Columns	14
Date coverage	01 Jan 2023 – 30 Jun 2025
Unique order IDs	1,200
Unique customers	1,189
Missing values after cleaning	0 across the 14 columns

The data-quality report associated with Project 1 records 309 blank coupon-code entries before cleaning. These were interpreted as orders with no coupon and replaced with "No Coupon". This is a reasonable working assumption, but it should be confirmed with the data owner if the dataset is used operationally.

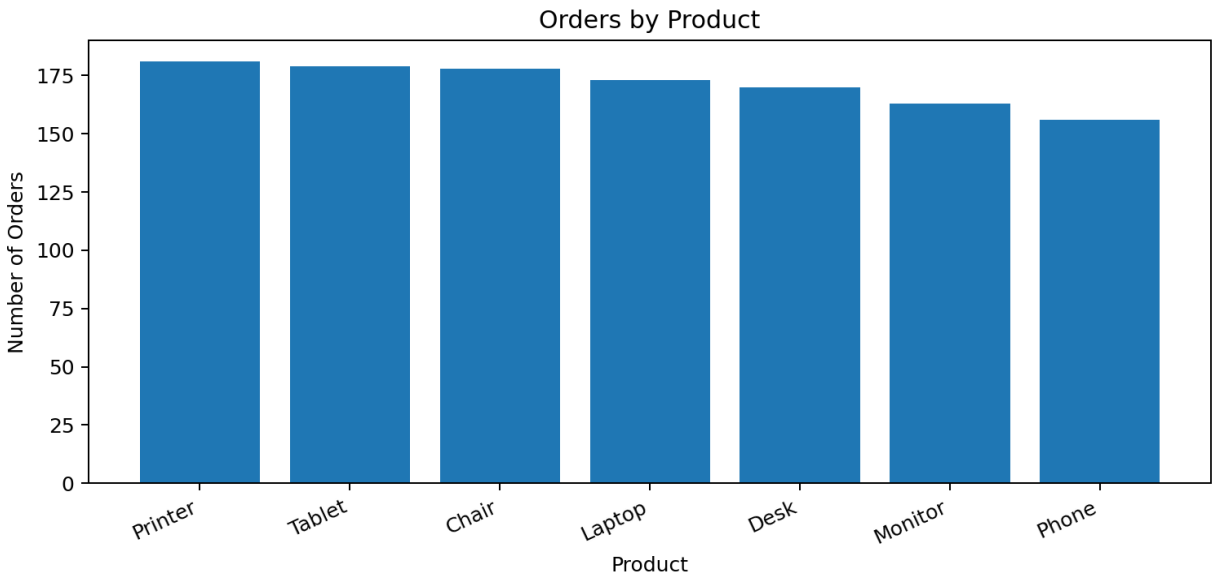
4. Descriptive Statistics

Descriptive statistics summarize the typical size and spread of order activity. The mean is the arithmetic average; the median is the middle value after sorting. The count tells us how many observations are included.

Measure	Result	Interpretation
Order count	1,200	There are 1,200 order records.
Total recorded order value	1,264,761.96	Sum of TotalPrice for every order status.
Mean order value	1,053.97	Average TotalPrice per order.
Median order value	823.62	Half of orders are at or below this value and half at or above.
Minimum order value	11.39	Smallest recorded TotalPrice.
Maximum order value	3,456.40	Largest recorded TotalPrice.
Mean quantity per order	2.95	Average quantity recorded per order.
Total quantity	3,535	Sum of the Quantity column.

The mean order value (1,053.97) is higher than the median (823.62). This indicates that higher-value orders pull the average upward. For describing a typical order, the median is less affected by unusually large values.

5. Product Analysis

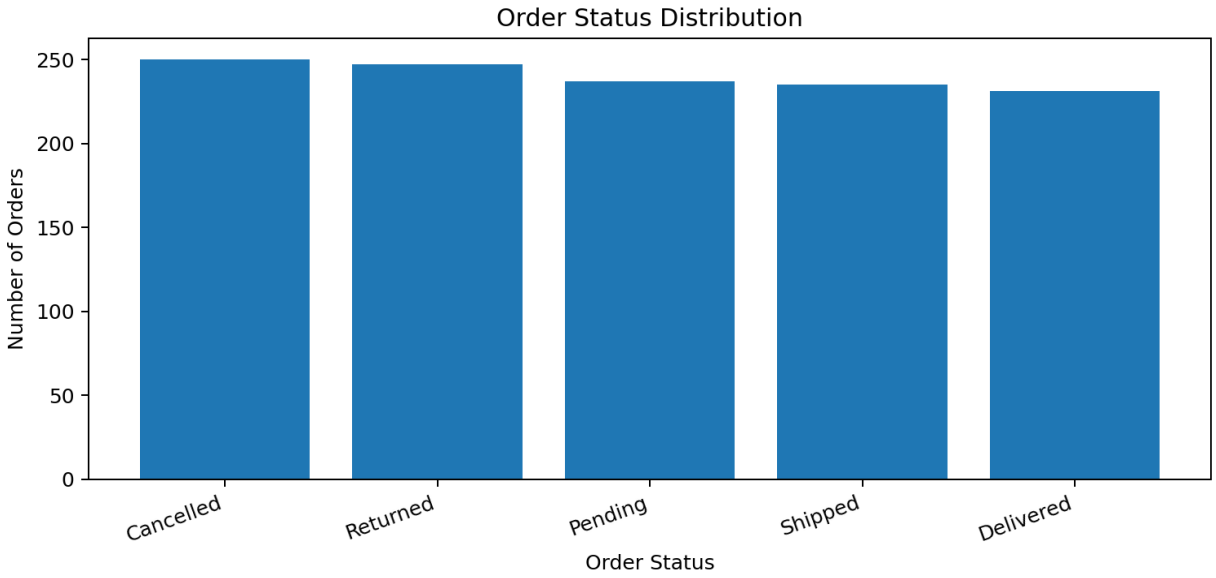


Printer has the highest order count (181), followed closely by Tablet (179) and Chair (178). Phone has the lowest count (156). The order counts are relatively close, so no single product dominates the dataset by volume. Chair has the largest summed recorded order value (195,620.11), narrowly above Printer (195,612.61). Phone has the lowest summed value (151,722.39).

Product	Orders	Quantity	Mean order value	Total order value
Printer	181	542	1,080.73	195,612.61
Tablet	179	497	1,042.28	186,568.95
Chair	178	562	1,098.99	195,620.11
Laptop	173	535	1,110.56	192,126.56

Product	Orders	Quantity	Mean order value	Total order value
Desk	170	508	985.06	167,459.93
Monitor	163	480	1,077.62	175,651.41
Phone	156	411	972.58	151,722.39

6. Order Status Analysis

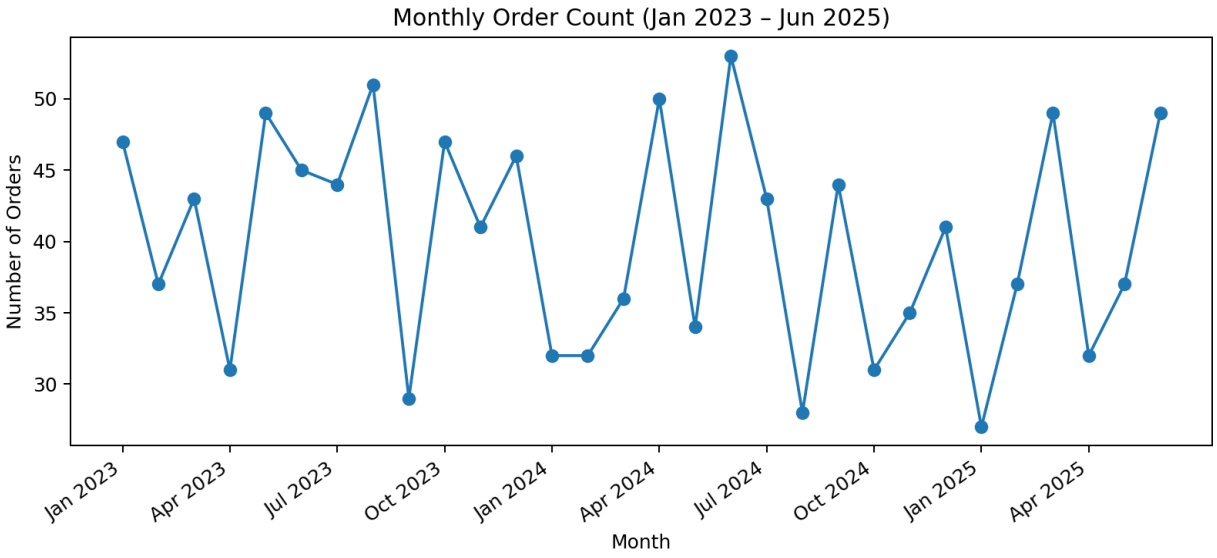


Cancelled is the most frequent status with 250 orders (20.8%), while Delivered is the least frequent with 231 orders (19.2%). The five statuses are distributed fairly evenly. Cancelled and Returned together account for 497 orders, or approximately 41.4% of the dataset. Pending accounts for a further 19.8%. This means that the status of an order must be considered before interpreting the full TotalPrice sum as sales or revenue.

Status	Orders	Share	Total recorded order value
Cancelled	250	20.8%	276,396.21
Returned	247	20.6%	243,277.70
Pending	237	19.8%	256,328.15
Shipped	235	19.6%	246,159.58
Delivered	231	19.2%	242,600.32

The dataset does not provide a documented business definition for when an order should count as revenue. Therefore, this report does not remove statuses or present a revenue figure. A business owner would need to define which statuses count as completed sales.

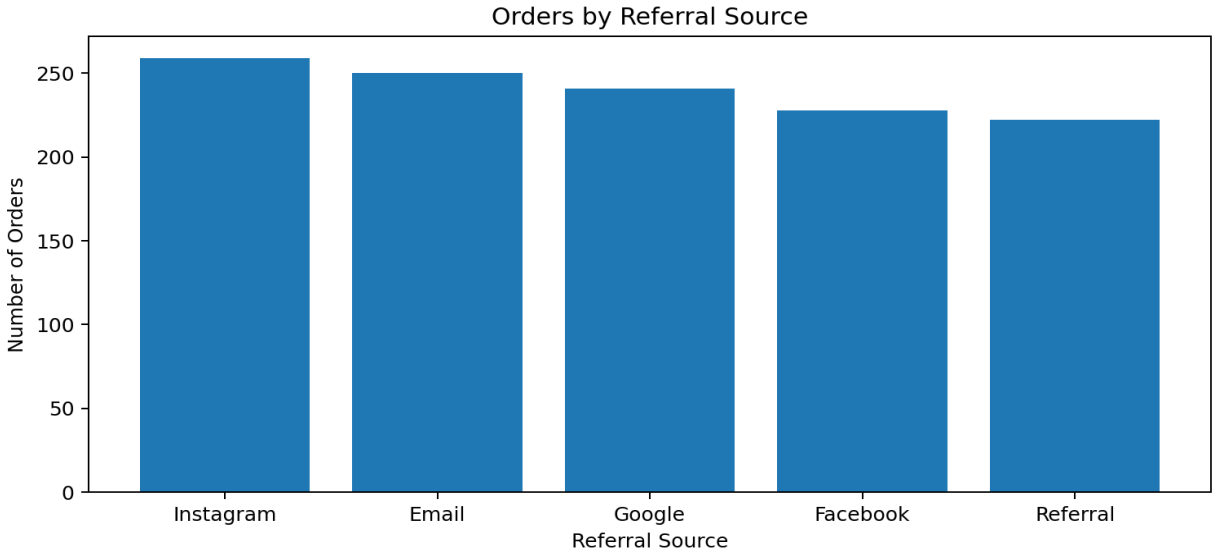
7. Time-Based Analysis



Monthly order counts fluctuate rather than following a steady upward trend. The highest monthly count is 53 orders in June 2024. The lowest is 27 orders in January 2025. June 2024 also has the highest monthly total recorded order value (68,068.54). The lowest monthly total recorded order value is 27,751.71 in April 2023.

The complete year 2024 has 459 orders compared with 510 in 2023, a decrease of 10.0%. The summed recorded order value decreases by approximately 13.1% from 552,643.24 in 2023 to 480,235.87 in 2024. The 2025 data covers only January through June, so it must not be compared directly with the full-year totals for 2023 and 2024.

8. Referral Source, Payment and Coupon Patterns



Instagram is the most common referral source with 259 orders, while Referral is the least common with 222. The difference is modest, suggesting that orders are spread across the five recorded sources. This is a count comparison only; it does not establish which channel is most profitable because marketing cost and completed-sale outcomes are not supplied.

Referral source	Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Payment methods are also distributed fairly evenly: Online (258), Cash (246), Credit Card (234), Debit Card (232) and Gift Card (230). Coupon categories are similarly balanced: FREESHIP (313), No Coupon (309), WINTER15 (292) and SAVE10 (286). These counts describe usage frequency, not the effect of a coupon on order value.

9. Outlier Detection

The Interquartile Range (IQR) method was used on TotalPrice. Q1 is 410.52 and Q3 is 1,578.475, giving an IQR of 1,167.955. The upper outlier fence is Q3 + 1.5 × IQR = 3,330.41. Eight orders are above this upper fence. There are no lower outliers because the lower fence is below zero and all recorded order values are positive.

These eight high values should be reviewed, not automatically deleted. The largest order is 3,456.40. All observed order totals are consistent with Quantity × UnitPrice (allowing for tiny floating-point precision differences), so the high values may be valid large-quantity / high-unit-price orders rather than data-entry errors.

10. Relationships Between Numerical Variables

The correlation between UnitPrice and TotalPrice is approximately 0.72, indicating a moderately strong positive linear relationship. Quantity and TotalPrice have a correlation of approximately 0.62. ItemsInCart and Quantity have a correlation of approximately 0.65. These associations are expected to some extent because order totals are calculated from quantity and unit price. Correlation does not prove causation.

11. Key Findings

1. The cleaned dataset is complete for the 14 supplied columns: no missing values remain, and all 1,200 Order IDs are unique.
2. The average order value (1,053.97) is greater than the median (823.62), showing that high-value orders increase the mean.
3. Printer records the highest number of orders (181), while Chair has the highest summed recorded order value (195,620.11).
4. Cancelled and Returned statuses together represent about 41.4% of orders, so status filtering is essential before reporting revenue.
5. June 2024 is the strongest month by order count (53) and recorded order value (68,068.54); January 2025 has the fewest orders (27).
6. Eight upper-end order-value outliers were detected by the IQR method; they should be investigated rather than removed automatically.
7. Referral, payment and coupon counts are relatively balanced across their categories.

12. Recommendations

• Review cancelled and returned orders to understand the reasons and operational impact.
• Define an approved rule for which order statuses count as completed sales before calculating revenue.
• Investigate the eight high-value orders to confirm they are valid transactions.
• Monitor monthly order counts and values, especially the decline from 2023 to 2024.
• Evaluate referral channels using completed orders, revenue rules and marketing costs rather than order counts alone.
• Confirm the meaning of blank coupon codes with the dataset owner.

13. Limitations

This analysis is descriptive and based only on the supplied dataset. It does not establish causes for monthly changes, customer behaviour or channel performance. The 2025 period is partial. No profit, discount amount, shipping cost, marketing spend or reason-for-cancellation fields are available. The interpretation of blank coupon codes is an assumption carried forward from Project 1. TotalPrice is treated as recorded order value, not recognized revenue.

14. Conclusion

Exploratory Data Analysis helped summarize the order dataset and reveal patterns that are not obvious from viewing individual rows. The analysis identified differences in product volumes, a fairly even distribution of order statuses, month-to-month fluctuations, and a small number of unusually high order values. It also highlighted an important business interpretation issue: the recorded order value includes orders with different statuses. Overall, EDA provides a foundation for more focused analysis and better business questions.

Appendix: Suggested Power BI Measures

```
Total Orders = COUNTROWS('Sheet1 (2)')
Total Recorded Order Value = SUM('Sheet1 (2)'[TotalPrice])
Average Order Value = AVERAGE('Sheet1 (2)'[TotalPrice])
Median Order Value = MEDIAN('Sheet1 (2)'[TotalPrice])
Total Quantity = SUM('Sheet1 (2)'[Quantity])
Unique Customers = DISTINCTCOUNT('Sheet1 (2)'[CustomerID])
```

If the table name in Power BI is Sheet1 rather than Sheet1 (2), replace the table name in each formula. The measure name “Total Recorded Order Value” is deliberate: do not rename it revenue unless the instructor or data owner confirms the correct status rule.